PYTHON DATA ANALYSIS – PROJECT REPORT

Project Title

Data Analysis Using NumPy and Pandas

Assignment: Python DA Assignment 1

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1. Introduction

Data analysis is the process of collecting, organizing, processing, and interpreting data to obtain meaningful information.

Python provides powerful libraries such as NumPy and Pandas that simplify numerical computations and data manipulation.

This project focuses on using NumPy and Pandas to perform basic data analysis operations, including array creation, numerical calculations, data filtering, indexing, grouping, and data manipulation.

The project is implemented using Jupyter Notebook.

2. Objectives of the Project

The main objectives of this project are:

- To understand the fundamentals of NumPy and Pandas.
- To create and manipulate one-dimensional and two-dimensional NumPy arrays.
- To perform mathematical operations on numerical data.
- To understand Pandas Series and custom indexing.
- To perform data selection, filtering, and slicing.
- To create and explore Pandas DataFrames.
- To perform data grouping and aggregation.
- To update, add, and delete records and columns.

3. Tools and Technologies Used

Tool / Technology	Purpose

Python	Programming language
Jupyter Notebook	Coding and execution environment
NumPy	Numerical computations and array operations
Pandas	Data manipulation and analysis

4. Project Implementation

The project is divided into three major sections.

4.1 NumPy Array Operations

Objective:

To understand how NumPy is used for creating arrays, inspecting array properties, performing mathematical operations, and extracting specific elements using indexing and slicing.

Tasks Performed:

1. Created a one-dimensional NumPy array named temperatures_w1 containing seven days of Week 1 temperature data.
2. Inspected the array's shape, data type, and number of elements.
3. Converted Celsius temperatures into Fahrenheit.
4. Calculated maximum, minimum, and mean temperatures.
5. Extracted the first three days, weekend temperatures, and middle three days using slicing.
6. Created a two-dimensional NumPy array named temperatures containing temperature data for two weeks.
7. Inspected the properties of the two-dimensional array.
8. Extracted weekly and weekend temperature data.

Temperature Dataset:

Week 1:
22.5, 25.3, 20.8, 23.4, 26.1, 24.8, 21.9

Week 2:
19.2, 22.5, 21.3, 24.0, 23.5, 22.8, 20.1

Key Results:

- Array dimensions: 1D and 2D.
- Week 1 maximum temperature: 26.1°C.
- Week 1 minimum temperature: 20.8°C.
- Week 1 average temperature: 23.54°C (approximately).
- Two-week array shape: (2, 7).
- Total elements in the two-week array: 14.

4.2 Pandas Series

Objective:

To understand Pandas Series creation, custom indexing, accessing values, filtering data, and manipulating Series elements.

Tasks Performed:

1. Created a Pandas Series named marks using student marks.
2. Assigned custom index labels from Rank1 to Rank5.
3. Accessed the first-rank student's mark using integer indexing.
4. Retrieved the top three ranks using the loc accessor.
5. Retrieved the third-rank student's mark using the iloc accessor.
6. Applied Boolean masking to retrieve students scoring above 90.
7. Modified the first-rank student's mark from 95 to 100.
8. Removed the last-rank student's entry.
9. Calculated CGPA by dividing each remaining mark by 10.

Student Marks Dataset:

Rank	**Marks**
Rank1	95

Rank2	92
Rank3	89
Rank4	85
Rank5	80

Key Results:

- Rank1 marks were updated to 100.
- Rank5 was removed successfully.
- Students scoring above 90 were identified.
- CGPA was calculated for the remaining four students.

Final CGPA Results:

Rank	Marks	CGPA
Rank1	100	10.0
Rank2	92	9.2
Rank3	89	8.9
Rank4	85	8.5

4.3 Pandas DataFrame

Objective:

To understand how to create, explore, filter, group, and manipulate structured transaction data using Pandas DataFrames.

Dataset Description:

A transaction dataset containing 10 records and four columns was created.

The dataset includes:

- TransactionID
- ProductCategory
- Region
- Amount

Tasks Performed:

A. DataFrame Creation

1. Created a Python dictionary containing transaction details.
2. Converted the dictionary into a Pandas DataFrame named transactions.
3. Displayed the complete transaction dataset.

B. Data Exploration

1. Displayed the first five records using head().
2. Displayed the last five records using tail().
3. Identified the DataFrame shape.
4. Displayed column names.
5. Inspected column data types and information using info().
6. Selected ProductCategory and Amount columns.
7. Retrieved the last three columns using iloc.
8. Filtered transactions where the Region was North and Amount was greater than 200.
9. Calculated category-wise transaction counts using value_counts().
10. Identified unique regions using unique().
11. Grouped transactions by Region and calculated the average transaction amount using groupby() and mean().

C. Data Manipulation

1. Updated the Amount of TransactionID 102 from 150 to 165.
2. Added a new Discount column by calculating 10% of the transaction amount.
3. Removed the row corresponding to TransactionID 109.
4. Deleted the Discount column from the DataFrame.

Key Results:

Product category counts:

Product Category	**Transactions**

Electronics	4
Clothing	3
Furniture	3

Average transaction amount by region (before manipulation):

Region	Average Amount
East	375.00
North	287.50
South	250.00
West	190.00

Additional results:

- TransactionID 102 was updated to 165.
- A 10% Discount column was created.
- TransactionID 109 was deleted.
- The Discount column was removed.
- Final DataFrame contains 9 rows and 4 columns.

5. Overall Results and Observations

The project successfully demonstrates the basic applications of NumPy and Pandas in data analysis.

The following observations were obtained:

- NumPy provides efficient operations for numerical arrays and mathematical calculations.
- Array slicing helps retrieve specific portions of numerical data.
- Pandas Series supports custom indexing and condition-based filtering.
- Pandas DataFrames make structured data exploration easier.
- Grouping and aggregation help identify regional transaction patterns.

- Data manipulation operations allow records and columns to be updated or removed efficiently.

6. Skills and Concepts Learned

Through this assignment, the following concepts were practised:

- Python library importing.
- NumPy array creation and manipulation.
- Array shape, size, and data type inspection.
- Mathematical and statistical operations.
- Array indexing and slicing.
- Pandas Series creation and manipulation.
- loc and iloc indexing.
- Boolean masking.
- DataFrame creation and exploration.
- Data filtering and column selection.
- Grouping and aggregation.
- Updating and deleting DataFrame records and columns.

7. Conclusion

This project provided practical knowledge of Python-based data analysis using NumPy and Pandas.

Through the three sections, numerical data was analysed using NumPy, student marks were managed using Pandas Series, and transaction records were explored and manipulated using Pandas DataFrames.

Overall, the project successfully covered the fundamental operations of NumPy and Pandas and demonstrated their importance in data analysis.

END OF PROJECT REPORT